

DIGIT: DIffusion-based Guitar Interactivity with Physics-informed Torque Control

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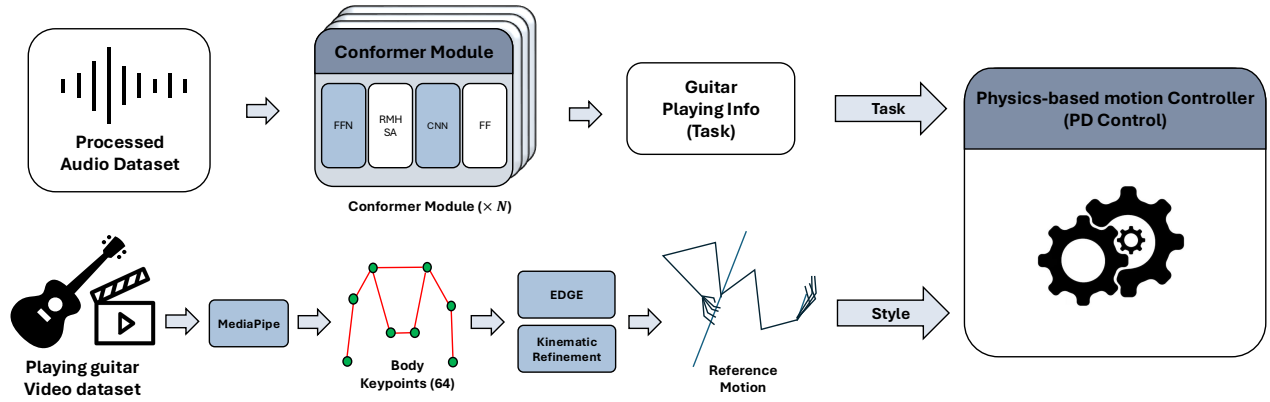


Fig. 1. 임시 흐름

Generating realistic guitar-playing motions is inherently challenging because guitar performance involves complex and fine-grained interactions between the human body and the instrument. Modeling such interactions would ideally require large-scale, high-quality motion capture data; however, in practice, available guitar performance data is extremely limited. Existing resources such as sheet music, MIDI (Musical Instrument Digital Interface), and 2D guitar videos provide only partial information: sheet music and MIDI lack performer-specific fingerings and hand motions, while video data suffer from motion blur, self-occlusion, and viewpoint limitations, making accurate finger-level motion recovery difficult.

To address these challenges, we introduce DIGIT, a physics-based framework that generates 3D guitar-playing motions from a single audio input. DIGIT consists of three stages: audio transcription, motion generation, and physics-based control. A Conformer-based transcription module extracts guitar tablature information, including string, fret, and timing, providing explicit guidance for precise finger placement. A diffusion-based motion generation module produces natural and expressive upper-body and finger motions, which are subsequently refined by a reinforcement learning-based control module to ensure physically plausible and musically accurate execution. We demonstrate that DIGIT can generate realistic and physically valid guitar-playing motions using audio input alone.

CCS Concepts: • **Computing methodologies** → **Motion processing; Neural networks; Animation; Adversarial learning; Physical simulation.**

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Additional Key Words and Phrases: Guitar performance, Audio-to-motion generation, Motion generation, reinforcement learning, adversarial imitation learning

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1 Introduction

With the rapid advancement of metaverse and VR/AR technologies, generating digital humans that not only resemble real people but also behave and interact naturally with their environment has become a central challenge in computer graphics and embodied AI. Among various human activities, musical performance represents one of the most expressive and skill-intensive forms of human behavior. In particular, synthesizing realistic instrument-playing animations holds significant potential for applications in virtual concerts, entertainment, and music education.

Despite recent progress in audio-driven motion generation, producing realistic instrument performance remains highly challenging. Unlike general human motion, musical performance involves complex and fine-grained interactions between the performer and the instrument. Guitar playing, in particular, requires asymmetric coordination between the two hands and precise, dexterous finger movements that must accurately contact specific strings and frets. Modeling such intricate interactions demands large-scale, high-quality motion capture data, yet guitar performance datasets with sufficient finger-level detail are extremely scarce.

Existing audio-based motion generation approaches have primarily focused on rhythmical full-body motions such as dance or

gestural behaviors, and are not well suited for instrument performance that requires continuous physical contact and precise finger control. When naively applied, generic generative models often produce physically implausible results, such as hands penetrating the instrument, which significantly degrades visual realism. Moreover, generating musically accurate guitar-playing motions requires explicit knowledge of which string and fret are played at each moment. However, raw audio signals do not directly encode such spatial or semantic performance information.

To overcome these challenges, we propose DIGIT, a unified framework that generates physically plausible and musically accurate 3D guitar-playing motions from a single audio input. DIGIT decomposes the problem into three stages: semantic audio transcription, motion generation, and physics-based control. First, a Conformer-based transcription model extracts guitar tablature information, string, fret, and timing, from the input audio, transforming ambiguous acoustic signals into explicit performance instructions. Second, a diffusion-based motion generation module synthesizes natural upper-body and finger motions that align with the musical rhythm and style, serving as a reference motion without enforcing strict physical constraints. Finally, a physics-based control module integrates reinforcement learning with adversarial motion priors to refine the generated motion in a simulation environment, ensuring accurate finger–guitar contact while respecting physical laws.

The main contributions of this work are as follows:

- (1) We introduce a Conformer-based transcription module that extracts fine-grained guitar performance semantics, including string, fret, and timing information, from polyphonic audio.
- (2) We present an integrated pipeline that combines diffusion-based motion generation with physics-based control to generate realistic guitar-playing motions under severe data scarcity.
- (3) We demonstrate that high-quality, viewpoint-independent 3D guitar-playing animations can be generated from audio alone, enabling applications in virtual performance and music education.

2 Related Work

Our work aims to generate physically plausible and musically accurate guitar-playing motions from audio input alone. We review related research from two perspectives: generative models for motion synthesis and physics-based character control, and clarify how our approach differs from recent instrument performance generation methods.

2.1 Generative Models for Motion Synthesis

Motion synthesis conditioned on semantic inputs such as text or audio has been extensively studied, with the goal of translating high-level signals into natural human movements. Early works, such as ERD [Fragkiadaki et al. 2015], primarily relied on recurrent neural networks including LSTMs and GRUs. However, these models are known to suffer from the mean prediction collapse problem [Martinez et al. 2017], where long-term motion predictions converge to an average pose, resulting in static or overly smooth motions.

Subsequent studies attempted to mitigate this issue by adopting more expressive architectures. For example, AIST [Li et al. 2021] introduced Transformer-based models for music-driven dance generation, enabling better temporal modeling over long musical sequences. Despite these improvements, generating diverse and expressive motions remained challenging.

More recently, diffusion-based models have emerged as a powerful paradigm for motion generation. Methods such as MDM [Tevet et al. 2022], EDGE [Tseng et al. 2022], and the work of Alexanderson et al. [Alexanderson et al. 2023] demonstrated that diffusion models can generate temporally coherent and stylistically diverse motions through probabilistic sampling. These properties make diffusion models particularly well suited for music-driven motion synthesis, where rhythmic consistency and subtle stylistic variations are essential.

However, most diffusion-based motion generation methods focus on free-space human motion and do not explicitly account for geometric or physical constraints. Instrument performance, especially guitar playing, imposes strict contact requirements between the fingers and specific locations on the instrument. As a result, diffusion models often produce physically implausible outcomes, such as fingers penetrating the instrument or failing to make proper contact.

To address such limitations, several works have explored hybrid approaches. Synchronize Dual Hands [Xu and Wang 2024a] employed reinforcement learning in a physics simulation to coordinate bimanual manipulation, while Canales et al. [Canales and Jörg 2025] combined procedural rules and inverse kinematics to generate guitar-playing motions from symbolic music scores. In particular, Canales et al. used inverse kinematics to stabilize RNN-generated motions and enforce finger placement on guitar strings. However, such kinematic approaches tend to restrict motion diversity and cannot fully resolve physical interaction errors, such as sliding or penetration, due to the lack of explicit physical modeling.

In contrast, our approach moves beyond reference-motion-dependent or purely kinematic solutions by directly incorporating explicit tablature information as a control target. This enables our framework to simultaneously achieve musical accuracy and physical plausibility within a unified system.

2.2 Physics-based Character Control

Physics-based character control has made significant strides, particularly through reinforcement learning frameworks such as DeepMimic [Peng et al. 2018] and AMP [Peng et al. 2021]. These methods enable simulated characters to perform complex behaviors while adhering to physical laws, allowing for realistic movement generation. Initially, DeepMimic [Peng et al. 2018] focused on tracking motion capture data to generate physically plausible movements, making it a foundational approach in the field. AMP [Peng et al. 2021] introduced adversarial learning, enabling the direct learning of complex motion styles from datasets without the need for detailed reward function design.

Early efforts in this domain primarily focused on basic locomotion tasks. However, recent advancements have expanded the scope of physics-based character control to more complex behaviors, such as

fine-grained manipulation and multi-modal interactions. ASE [Peng et al. 2022] and CALM [Tessler et al. 2023], for example, utilized large-scale motion data to train agents to learn low-level control skills in latent spaces, allowing for the generation of flexible and contextually appropriate motions.

As these advances in character control progressed, there has been an increasing emphasis on replicating realistic human movements by modeling human-environment interactions. In the real world, human actions are often driven by interactions with objects and the environment, whether through visual or tactile engagement. Thus, accurately replicating human movement necessitates the incorporation of these interactions. Recent studies have attempted to model human-environment interactions to enhance the realism of simulated characters [Bae et al. 2023; Hassan et al. 2023; Lee et al. 2023; Li and Yu 2023].

For instance, Hassan et al. [Hassan et al. 2023] leveraged motion capture data that included both human movements and interactions with objects, enabling virtual characters to interact realistically with objects such as chairs, sofas, and tables. However, their approach is constrained by the limited set of objects in the dataset, restricting the range of possible interactions (e.g., sitting, lying down, carrying boxes). In contrast, Lee et al. [Lee et al. 2023] reproduced full-body human movements using minimal sensors, such as headsets and hand controllers. Although this approach reduces the need for extensive tracking, it still relies on motion databases to estimate untracked body movements, limiting the realism of the generated motions and the sensor coverage.

While these studies demonstrate the potential of human-environment interaction modeling, they are still heavily reliant on motion datasets that include only a specific set of interactions. This results in a limited ability to reproduce diverse interactions with a variety of objects and environments. Furthermore, because this research is still in its early stages, the scarcity of comprehensive interaction datasets presents a significant challenge in achieving fully realistic human-object interactions.

Despite these advancements, significant challenges remain, particularly in tasks that require fine-grained two-handed manipulation, complex object interactions, and the integration of timing and rhythm, as in musical instrument performance. These tasks involve a high degree of coordination between multiple factors, making them much more complex than simple locomotion or object manipulation.

To address these challenges, this research adopts the AMP framework to simulate guitar playing. While guitar playing involves interaction with an object (the guitar), it also introduces unique complexities that are not fully addressed by existing object interaction models, such as those proposed by Hassan et al. [Hassan et al. 2023]. These models primarily focus on static object interactions, which are insufficient for capturing the dynamic nature of musical instrument performance. Furthermore, manipulation-centric models like UniDexGrasp [Xu et al. 2023], which focus on precise object manipulation, tend to overlook the broader rhythmic movements and stylistic expression of a full-body performer. Therefore, AMP is chosen for its ability to handle dynamic, multi-faceted tasks, making it better suited to capture the nuances of guitar playing.

2.3 Motion Synthesis for Musical Instrument Performance

Early research on musical instrument performance animation has primarily relied on rule-based systems [ElKoura and Singh 2003; Zhu et al. 2013] or supervised learning frameworks that map audio and MIDI signals to motion [Chen et al. 2021; Kao and Su 2020; Li et al. 2018; Shlizerman et al. 2018]. However, these approaches have limitations in capturing the intricate interactions between the performer and the instrument, often resulting in physically implausible motions, such as joint penetration.

The most closely related works to ours are recent studies on physics-based instrument performance, including Xu et al. [Xu and Wang 2024b] for guitar playing and ELGAR [Qiu et al. 2025] for cello performance. These works address the penetration and contact issues inherent in kinematic approaches by introducing physical constraints through reinforcement learning or physics-aware loss functions.

Xu et al. [Xu and Wang 2024b] proposed a reinforcement learning (RL) framework for generating synchronized bimanual guitar-playing motions in a physics-based simulator. Their method follows a two-step pipeline: first, a data-driven model generates kinematic reference motions from audio or MIDI input. Then, an RL agent learns to track these motions in the simulator, mapping high-level kinematic targets to low-level joint torques, ensuring realistic hand-instrument contact and synchronized movements. This approach effectively addresses common issues in purely kinematic methods, such as self-penetration and inaccurate contact, by incorporating a physics-based simulator to enforce physical plausibility. Despite its strengths, the framework heavily relies on the quality of reference motions, which are generated from audio or MIDI data. Furthermore, although the use of symbolic score information (e.g., tablature) helps integrate musical content into the training process, the framework still faces challenges in musical expressiveness and stylistic diversity. The RL agent focuses primarily on motion tracking, and while it ensures physical feasibility, it does not explicitly incorporate higher-level musical nuances such as timing variations, dynamics, or emotional expression in the performance. Our work builds upon the effectiveness of physics-based control while further enhancing musical precision by explicitly incorporating tablature-level targets—string and fret information—directly into the control objective.

ELGAR [Qiu et al. 2025] demonstrated high-quality cello performance generation using a diffusion-based model augmented with physics-inspired contact losses. This hybrid approach highlights the benefit of combining the expressiveness of generative models with physical constraints. Inspired by this line of work, we adopt a complementary strategy: we first generate expressive motions using a diffusion model, refine them kinematically, and then use them as reference motions for physics-based control in simulation.

In summary, our framework inherits the strengths of both diffusion-based motion generation and reinforcement learning-based physical control, while introducing tablature-guided precision tailored specifically to guitar performance. This combination enables us to generate motions that are not only physically valid but also musically accurate at the finger-instrument interaction level.

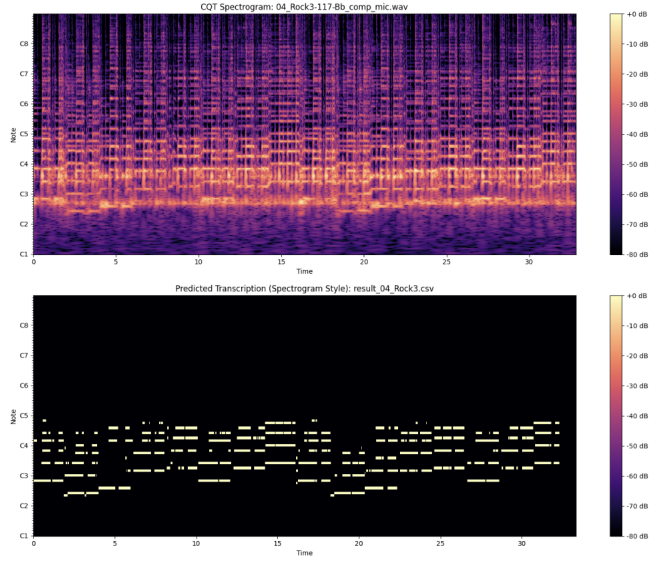


Fig. 2. Visualization of the audio-to-tablature transcription process, illustrating the transformation from a CQT spectrogram to a structured tablature representation. The top figure shows the CQT spectrogram of a guitar audio signal containing complex harmonic structures and noise. The bottom figure depicts the guitar tablature predicted by the Conformer model. The model effectively suppresses harmonic interference and reconstructs discrete note events, including pitch, onset, and offset.

3 Overview

The goal of this work is to generate realistic guitar performance motions from audio, accurately capturing the interactions between the performer and the instrument. To achieve this, it is first necessary to extract tablature information, which encodes string and fret actions, from the input audio. We train a Conformer-based model for this task using several preprocessed guitar audio datasets [Kehling et al. 2023; Pedroza Villalobos et al. 2025; Xi et al. 2019], mapping audio signals directly to tablature sequences. In addition to tablature extraction, generating natural upper-body and finger motions is essential. To this end, 3D motions are extracted from guitar performance videos and used to train a diffusion-based motion generation model, adapted from EDGE (Editable Dance GEneration) [Tseng et al. 2022]. Finally, a control policy is learned in a physics-based simulation environment using the Adversarial Motion Priors (AMP) framework [?]. We design the reward function to encourage the policy to faithfully follow both the extracted tablature and the reference motion generated in the previous stage, ensuring physically plausible and musically accurate performance. An overview of our complete framework is illustrated in Figure 1.

4 Audio-to-Tablature Transcription

Generating physically plausible 3D guitar performance from a single audio signal requires more than recognizing musical content; it demands explicit information about how the sound is produced on the instrument. A guitar comprises six strings and multiple frets, and identical pitches can be realized through different string–fret combinations, each inducing distinct hand positions and fingering

configurations. Consequently, accurate motion synthesis necessitates a tablature-level representation that specifies which pitch is played, on which string, at what time, and for what duration.

To extract such representations from audio, we construct an audio-to-tablature transcription model trained on a unified dataset composed of GuitarSet [Xi et al. 2019], IDMT [Kehling et al. 2023], and guitar_techs [Pedroza Villalobos et al. 2025]. The combined dataset contains approximately 560 minutes (9.3 hours) of audio recordings with aligned annotations. GuitarSet [Xi et al. 2019] provides fine-grained labels for a wide range of acoustic guitar techniques, IDMT [Kehling et al. 2023] contributes diversity in guitar types and recording conditions, and guitar_techs [Pedroza Villalobos et al. 2025] captures expressive playing styles characteristic of electric guitars. To ensure compatibility across datasets, we convert all annotations into a common format encoding note onset time, offset time, string index, and pitch.

As audio input, we use Constant-Q Transform (CQT) spectrograms computed on a logarithmic frequency scale to better reflect human auditory perception. All recordings are resampled to 22,050 Hz prior to CQT computation. The resulting time–frequency representations serve as inputs to the transcription network.

We adopt a Conformer architecture [Gulati et al. 2020] for tablature prediction. While Transformer-based models effectively capture long-range temporal dependencies, they often struggle with local acoustic details that are critical for distinguishing guitar articulations. The Conformer mitigates this limitation by combining self-attention with convolutional modules, enabling simultaneous modeling of short-term spectral patterns and long-term musical structure. This hybrid design is well suited for guitar audio, where transient events such as note attacks coexist with globally coherent playing sequences.

Figure 2 visualizes the transcription process. Given an input CQT spectrogram with densely overlapping frequency components (top), the model outputs a structured and denoised representation of discrete note activations (bottom). The model produces a $T \times 6$ matrix, where each element represents the predicted pitch for a given time frame and string. We post-process this matrix into a set of discrete note events of the form $[start, end, string, pitch]$. This explicit tablature representation serves as a critical interface between audio analysis and motion synthesis. By providing precise string and pitch information over time, it imposes geometric constraints on finger placement in the subsequent physics-based controller. As a result, the system is guided toward physically executable finger–fret interactions, enabling realistic guitar playing motions beyond appearance-level imitation.

5 Motion Generation

While tablature provides explicit information about which strings and frets should be activated over time, it does not uniquely define how the body should move to realize the performance. Mapping discrete string–fret targets to full-body joint trajectories constitutes a highly underdetermined control problem, especially when physical constraints such as joint limits, self-collisions, and contact dynamics are considered. Consequently, additional priors on natural human



Fig. 3. Upper-body and hand joint estimation from guitar performance videos using the MediaPipe Holistic model [Lugaresi et al. 2019]. We estimate 22 upper-body joints and 42 hand joints from multiple performance videos, while excluding facial and lower-body joints that are not required for motion generation.

motion are required to guide the synthesis toward realistic and physically plausible guitar playing behaviors.

5.1 Diffusion-Based Motion Generation from Audio

To capture natural guitar performance styles, we leverage a dataset of 2D guitar performance videos. Prior works such as DualHand [Xu and Wang 2024a] and Canales et al. [Canales and Jörg 2025] focus primarily on generating finger motions in isolation; however, such approaches often result in unnatural shoulder or elbow movements. In contrast, guitar performance inherently involves coordinated upper-body motion, including rhythmic torso sway and arm movement. We therefore aim to generate full upper-body motion rather than finger motion alone.

Due to frequent occlusions caused by the instrument during guitar performance, we adopt the MediaPipe Holistic pipeline [Lugaresi et al. 2019], which is robust to partial body visibility. MediaPipe Holistic combines the BlazePose full-body pose estimator [Bazarevsky et al. 2020] with the MediaPipe Hands model [Zhang et al. 2020].

아래에서 joint 개수 설명한 거랑 숫자가 다른 거에서 혼동이 올 수 있을 것 같아, 자세한 숫자는 제외하고 작성하였습니다. Using this framework, we extract 3D joint trajectories encompassing the upper body and detailed finger movements. Examples of the extracted joint configurations are shown in Figure 3. 3.

수빈 260121.

To facilitate motion control, these keypoints are mapped onto the SMPL skeleton format with 52 joints, supplemented by 10 additional fingertip positions to ensure precise finger-string interaction. These position-based trajectories are then converted into 52 joint rotation parameters. For the final training process, we utilize a subset of 43 joints, comprising the full upper-body and hands while excluding the lower body to focus on performance-relevant dynamics.

For motion generation, we adopt EDGE (Editable Dance GEneration) [Tseng et al. 2022], a diffusion-based framework that predicts

temporally coherent body motion from audio features using a Transformer architecture. We redesign the input representation and training procedure to accommodate guitar performance data. By using the preprocessed 43-joint upper-body and hand motion sequences as training targets, the model learns both fine-grained finger articulations and global upper-body rhythmic patterns characteristic of guitar playing. The output of the model is a pose sequence of a full-body skeleton over N frames. full-body면 학습에 사용하는 joint가 64개가 아닌가요? 총 몇 개인거죠? The model outputs a 3D pose sequence of size $64 \times 3 \times N$, representing 64 joints in three spatial dimensions over N frames.

수빈 260121.

학습에 사용되는 joint는 43개입니다..! 그렇게 되는 이유를 문단 사이에 추가하여 작성하였고, 위 문단에서 마지막 문장을 아래와 같이 수정하면 될 것 같습니다.

The output of the model is a sequence of joint rotations for the 43 upper-body and hand joints over N frames. This representation allows the generated motion to be directly mapped onto our physical character for simulation.

유리 260118.

EDGE 모델 학습을 위해서 사용한 video dataset에 대한 specification을 적어야 할 것 같습니다. 비디오 데이터는 어디서 가져온 건지(출처, source가 유튜브?), 총 사용한 비디오 개수와 음악 종류 (acoustic only), 총 영상 데이터셋 길이 (총 몇 분인지) 그리고 비디오 데이터 selection 기준. (정면에서 상체가 잘 나온 기타 연주 영상을 선택했다 이런 식으로)

그리고 output은 매 프레임마다 64개 joint position이 나오는 건가요? 이 결과를 retargeting해서 physics-simulation에서 사용하는 character에 retargeting하는 거라고 이해했는데 맞는지 확인해주세요. 제가 이해한 바를 다시 정리하자면, 비디오에서 estimation한 pose (keypoint position 형태)를 사용해서 EDGE 모델을 학습합니다. 학습된 EDGE 모델은 inference 시에 audio를 넣으면 그에 맞는 N frame 길이의 pose sequence(keypoint position 형태)가 나옵니다. 하지만 이 pose sequence는 기타와의 interaction을 고려하지 않은 상태이기 때문에 kinematic refinement가 필요합니다. 그리고 마지막으로 retargeting을 사용해서 우리가 control할 character에 모션데이터 (joint angle 형태의 모션)를 맞춥니다. 제가 제대로 이해한 게 맞을까요?

수빈 260121.

We curated a dataset of 11 high-quality guitar performance videos from YouTube, totaling approximately 38 minutes. Videos were selected based on the clear visibility of the performer's upper body and the guitar's frontal view to ensure robust 3D pose estimation. These datasets may be added more.

Using the MediaPipe Holistic pipeline, we extracted 3D keypoints encompassing both full-body poses and detailed hand articulations. These keypoints were then mapped onto the SMPL skeleton format. To capture precise finger-string interactions, we augmented the dataset with additional fingertip position data.

추가) 비디오에서 MediaPipe로 추출한 output은 전신(33) + 손(21+21) 해서 75개의 position data인데, 그대로 학습에 사용하지 않고 물리 시뮬레이션 캐릭터 제어에 적합하도록 SMPL 회전 데이터로 변환하였습니다. 이 과정에서 연주와 직접적인 관계가 없는 하반신을 제외하여, 최종적으로는 상반신과 손에 집중하여 총 43개의 joint를 EDGE 모델의 입력으로 사용하였습니다.

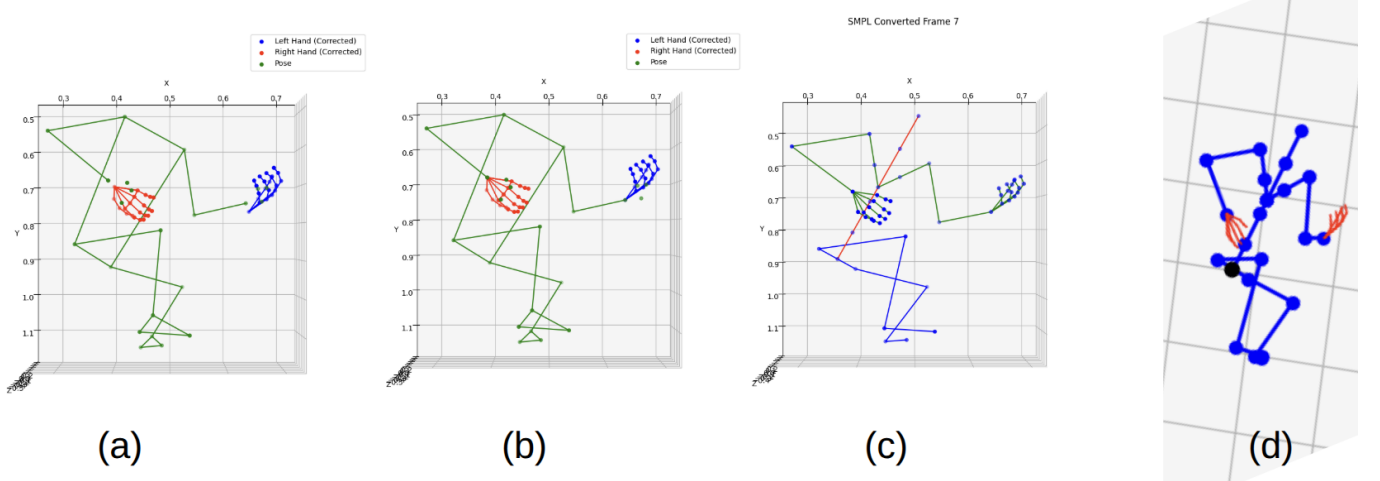


Fig. 4. (a) 입력된 RGB 영상에서 MediaPipe Holistic을 이용해 몸통과 양손의 키포인트를 개별적으로 추출한다. (b) 개별 좌표계로 추정된 몸통과 손의 키포인트를 단일 월드 좌표계로 통합하고, 골반(Pelvis)을 원점으로 하여 정렬한다. (c) 소스 골격과 타겟 골격 간의 구조적 차이를 해결하기 위해 관절 매핑 및 벡터 기반 구조 변환 (d) 최종적으로 분석적 역운동학 및 3-Point Orientation 기법을 적용하여 3차원 위치 정보를 6자유도 회전 정보로 변환한다. 이미지는 추후 통일성 있게 수정하겠습니다.

5.2 Kinematic Refinement

유리 260118.

Kinematic Refinement 어떻게 했는지 잘 모르겠습니다. 자세히 적어주면 좋겠어요. 그리고 **kinematic refinement**과 **motion retargeting** 선후 관계, 뭐가 먼저인지가 궁금합니다.

Although the generated motion captures rhythmic upper-body movement and arm trajectories, it does not explicitly account for physical interactions with the guitar, such as string contact or fret pressing. Therefore, the synthesized motion is not used directly as the final performance. Instead, it undergoes a kinematic refinement stage and serves as a reference motion for the subsequent physics-based control module. This reference motion provides a strong prior on natural human movement while allowing the controller to enforce physical constraints required for realistic guitar playing.

The raw joint coordinates obtained from guitar performance videos are dependent on the performer's body proportions and camera viewpoint, and their skeletal structure does not directly correspond to that of the physics-simulated agent used in our system. To resolve this discrepancy, we perform a motion retargeting process, as illustrated in Fig. 4. We first align coordinate systems using the pelvis as a reference and map joints between source and target skeletons. Directional vectors from the source motion are then transferred to the target model using a vector-based transformation. Finally, analytical inverse kinematics is applied to convert 3D joint positions into joint rotations, yielding kinematically consistent motions suitable for the simulation environment.

6 Phycis-based Control

260107 규칙

EDGE 모델을 통해 생성된 모션(M_{raw})은 음악적 리듬과 스타일을 반영하고 있지만, 기타라는 객체와의 기하학적 상호작용을 고려하지 않고 생성된 자유 공간 상의 움직임이다. 이 모션을 그대로 물리

시뮬레이션의 참조 모션으로 사용할 경우, 손이 악기 내부로 진입하려는 기하학적 모순으로 인해 기타 객체의 위치나 회전 상태가 불안정해지거나, 손가락이 줄에 닿지 않는 등의 문제가 발생하여 강화학습의 수렴을 저해한다. 따라서 본 연구에서는 물리 제어 단계 이전에, **Tablature** 정보를 활용한 역운동학 기반의 동작 정제 과정을 수행한다. 우선, 기타와 연주자 간의 상대적인 위치 관계를 정의하기 위해, 가상의 기타 객체를 이상적인 기타 연주 시 예상되는 위치에 배치하여, 상체의 움직임에 따라 기타가 자연스럽게 동기화되도록 설정한다. 이후, 다음의 두 단계를 거쳐 손의 궤적을 수정한다. **Tablature-guided Target Projection**: 오디오 분석 모듈(3.1절)에서 추출된 타블라추어 정보 $T_t = (s_t, f_t)$ 를 바탕으로, 매 프레임 t 마다 손이 위치해야 할 3D 목표 좌표 p_{target} 를 계산한다. 미리 정의된 기타의 기하학적 모델을 통해 s 번째 줄과 f 번째 프렛의 교차점 좌표를 획득하며, 이는 왼손의 운지 목표가 된다. 오른손의 경우, 스트로크 혹은 피킹 동작을 위해 해당 줄 s 의 높이를 목표 평면으로 설정한다. **IK-based Correction & Blending**: 계산된 목표 좌표 p_{target} 을 향해 분석적 IK 솔버를 적용하여 수정된 팔의 관절 각도 $\hat{q}_{ik}$ 를 산출한다. 이때, IK에 의해 동작이 지나치게 경직되는 것을 방지하고 **Diffusion** 모델이 생성한 고유의 스타일을 보존하기 위해, 원본 모션의 관절 각도 q_{raw} 와 IK로 수정된 각도 $\hat{q}_{ik}$ 를 선형 보간하여 최종 참조 모션 q_{ref} 를 생성한다.

$$q_{ref} = (1 - \lambda) \cdot q_{raw} + \lambda \cdot \hat{q}_{ik} \quad (1)$$

여기서 λ 는 보정 강도를 조절하는 가중치이다. 이렇게 정제된 모션 q_{ref} 는 물리적으로 타당한 손-악기 상호작용을 포함하게 되며, 이어지는 물리 제어기(3.4절)의 AMP 학습 과정에서 에이전트가 모방해야 할 고품질의 가이드 역할을 수행한다.

6.1 Physical-based motion control

260108 규칙

본 연구는 물리 시뮬레이션 환경 내에서 물리 법칙을 준수하면서도, 실제 연주자의 고유한 동역학적 스타일을 모방하는 자연스러운 기타 연주 동작 생성을 목표로 한다. 이를 위해 우리는 Adversarial Motion Priors (AMP)[?] 프레임워크를 채택하여, 에이전트가 수집된 데이터셋 으로부터 추출된 모션 패턴을 학습하도록 유도한다. 제어기는 상위 정책이 생성한 추상적인 행동 목표를 물리 시뮬레이션 상의 실제 움직임으로 변환하는 실행기 역할을 수행한다. 구체적으로, 정책 $\pi(a_t|s_t)$ 는 현재 상태 s_t 에서 스타일과 태스크를 만족시키기 위한 최적의 목표 관절 각도($\hat{q}$)를 행동 a_t 로 출력한다. 본 시스템은 안정적인 동작 제어를 위해 PD 제어기(Proportional-Derivative Controller)를 사용한다. PD 제어기는 목표 각도 $\hat{q}$ 와 현재 캐릭터의 관절 각도 q 및 각속도 $\dot{q}$ 간의 오차를 기반으로, 각 관절 구동에 필요한 토크(Torque, τ)를 다음 식과 같이 계산한다:

$$\tau = k_p(\hat{q} - q) - k_d\dot{q} \quad (2)$$

여기서 k_p 와 k_d 는 각각 관절의 강성과 감쇠 계수를 의미한다. 계산된 토크 τ 는 시뮬레이션 내의 물리 엔진에 전달되어 캐릭터를 구동시킨다. 결과적으로 정책이 의도한 연주 스타일과 정확한 타건 위치는 PD 제어기에 의해 물리적으로 타당한 힘으로 변환됨으로써, 떨림이나 미끄러짐 없는 안정적인 연주 동작으로 구현된다. 본 연구의 핵심 목표는 특정 시점에 정확한 프렛과 줄을 타건하여 명확한 연주음을 생성하는 것이다. 이를 위해 총 보상 r_t 는 스타일 보상 r_{style} 와 태스크 보상 r_{task} 의 가중 합으로 정의된다. **Fingering Reward (r_{finger})**: 실제 기타 연주에서 선명한 음향을 얻기 위해서는 손가락 끝이 프렛 와이어 사이의 중앙부를 누르는 것이 중요하다. 가장자리부를 누를 경우 버징 현상이 발생하거나 소리가 묵음 처리될 수 있기 때문이다. 따라서 우리는 목표 프렛의 시작 좌표와 끝 좌표로 정의되는 유효 구간 내에서, 손끝 위치 p_{tip} 이 프렛의 중심 p_{center} 에 가까울수록 가우시안 분포에 따른 높은 보상을 부여하도록 설계하였다. 반면, 해당 유효 구간을 벗어날 경우 급격한 페널티를 부여하여 에이전트가 정확한 프렛 범위를 엄수하도록 강제한다. 이를 수식으로 나타내면 다음과 같다:

$$r_{finger} = \exp\left(-\frac{|p_{tip} - p_{center}|^2}{2\sigma_{finger}^2}\right) \cdot \mathbb{I}(p_{tip} \in \mathcal{R}_{fret}) \quad (3)$$

여기서 σ_{finger} 는 허용 오차를 조절하는 표준편차이며, $\mathbb{I}(\cdot)$ 는 손끝이 목표 프렛 영역 $\mathcal{R}_{fret}$ 내에 존재할 때만 1을 반환하는 지시 함수이다. 에이전트는 이 보상을 최대화하기 위해 손끝을 유효 구간 내의 최적 위치로 향하게 하며, 이 과정에서 손끝과 연결된 상위 관절들의 구성은 자연스럽게 최적화된다.

Plucking Reward (r_{pluck}):

왼손이 음의 높낮이를 결정한다면, 오른손은 정확한 타이밍에 줄을 진동시켜 소리의 시작을 만드는 역할을 수행한다. 따라서 오른손 보상은 에이전트가 연주해야 할 특정 줄에 접근하도록 유도하는 거리 항과, 실제 물리적 접촉이 발생했을 때 부여하는 충돌 항으로 구성된다. 우선, 타블라추어 정보에 따라 현재 타격해야 할 줄의 위치 p_{string} 과 오른손 피킹 지점 p_{pick} 사이의 거리를 최소화해야 한다. 이때, 단순히 손이 줄에 닿아있는 것을 넘어, 줄을 뚫고 지나가는 타격 동작을 유도하기 위해, 우리는 줄의 위치를 기준으로 한 좁은 타격 유효 범위를 설정하였다. 수식은 다음과 같다:

$$r_{pluck} = w_{dist} \cdot \exp(-\beta|p_{pick} - p_{string}|^2) + w_{hit} \cdot \mathbb{I}_{contact} \quad (4)$$

여기서 첫 번째 항은 손이 목표 줄에 근접하도록 유도하는 가우시안 보상이다. 두 번째 항의 $\mathbb{I}_{contact}$ 는 물리 엔진 상에서 오른손과 목표 줄 사이에 실제 충돌이 감지되었을 때 1을 반환하는 지시 함수이며, w_{dist} 와 w_{hit} 는 각 항의 가중치이다. 특히 $\mathbb{I}_{contact}$ 항은 Tablature 상의 노트 발생 시점 전후의 짧은 시간 간격 내에서만 활성화되도록 설계하여, 불필요한 잡음을 생성하는 접촉을 방지하고 리듬에 맞는 정확한 타격을 강제한다.

Guitar Holding Reward (r_{hold}):

기타 연주 중 격렬한 동작으로 인해 악기가 캐릭터의 제어 범위를 벗어나거나, 연주자의 상체 움직임과 따로 노는 부자연스러운 현상을 방지해야 한다. 실제 연주 시, 기타는 양 손이나 스트랩 등을 통해 연주자의 상체에 밀착되어 있으며, 몸통의 회전이나 기울임에 따라 자연스럽게 동기화 되어 움직인다. 이를 구현하기 위해, 우리는 3.3절에서 정의한 바와 같이 캐릭터의 척추 관절 좌표계에 고정된 가상의 목표 기타를 설정하였다. 이 가상 기타는 상체의 움직임에 따라 이상적인 연주 위치를 실시간으로 갱신한다. 보상 함수는 물리 시뮬레이션 상의 실제 기타 위치 p_{sim} 이 이 목표 위치 p_{target} 으로부터 이탈하는 것을 최소화하도록 설계되었다:

$$r_{hold} = \exp(-\alpha|p_{sim} - p_{target}|^2) \quad (5)$$

여기서 p_{sim} 은 시뮬레이션 내 기타의 현재 중심 좌표이며, p_{target} 은 상체 모션에 동기화된 가상의 목표 좌표이다. α 는 위치 오차에 대한 민감도 계수이다. 이 함수는 에이전트가 역동적인 연주 동작을 수행하는 동안에도, 악기를 신체의 적절한 위치에 견고하게 파지하고 상체와 일체감 있게 움직이도록 유도한다.

7 Experiment

7.1 Experimental Setup

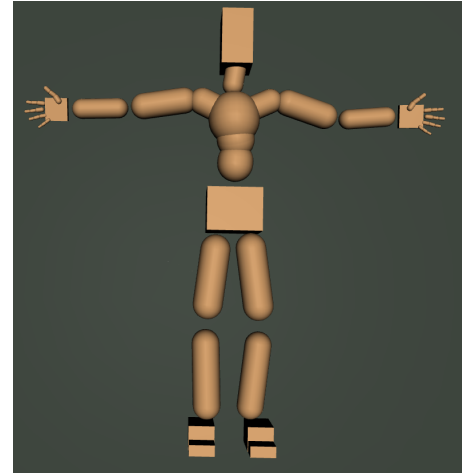


Fig. 5. SMPL-H[Romero et al. 2017] 기반 물리 시뮬레이션 모델. 각 손가락 마디가 관절로 구현된 고자유도 모델이다.

유리 251229.

To generate dexterous guitar playing motions, we employ a physics-based simulation model built upon the SMPL-H human body representation[Romero et al. 2017].

기존의 표준 휴머노이드 모델들은 연산 효율성을 위해 손을 하나의 덩어리로 취급하거나, 손바닥과 손가락을 잇는 단일 관절 형태로 단순화하는 경우가 일반적이다[Merel et al. 2017; Peng et al.

2018]. 반면, Figure 5에서 볼 수 있듯, 본 연구의 모델은 실제 인체와 유사한 다관절 손가락 구조를 온전히 포함하고 있어, 운지법과 같은 정교한 기타 주법을 물리적으로 재현하는 데 적합하다.

260105 기훈

우리는 인터넷에 공개된 3D 기타 urdf 모델을 활용하였으며, 각 스트링마다 fret의 시작점과 끝점에 시각적으로 렌더링되지 않는 가상 구형 조인트를 배치하여 이를 기준점 P_{start} 와 P_{end} 로 설정하였다. 본 실험에서는 총 $N = 10$ 개의 fret 구간으로 나누었으며, 전체 스트링 벡터를 N 등분한 ‘기준 단위 벡터(Base Unit Vector)’ $\mathbf{d}_{base}$ 를 다음과 같이 정의한다.

$$\mathbf{d}_{base} = \frac{P_{end} - P_{start}}{N} \quad (6)$$

각 fret의 구간 길이는 이 기준 벡터에 사용자 정의 비율 벡터 S 를 적용하여 결정된다. j 번째 fret($j = 1, \dots, N$)의 구간 벡터 $\Delta \mathbf{p}_j$ 는 다음과 같다.

$$\Delta \mathbf{p}_j = \mathbf{d}_{base} \cdot S_j \quad (7)$$

여기서 S_j 는 스케일 비율 벡터 S 의 j 번째 요소이다. 최종적으로 에이전트가 타건해야 할 j 번째 fret의 목표 중심 좌표(Target Midpoint) M_j 는 누적된 시작 위치에서 해당 구간 벡터의 절반만큼 이동한 지점으로 계산된다.

$$M_j = P_{j-1} + 0.5 \cdot \Delta \mathbf{p}_j, \quad \text{where } P_j = P_{j-1} + \Delta \mathbf{p}_j \text{ and } P_0 = P_{start} \quad (8)$$

본 실험에서 사용된 비율 벡터 S 는 실제 기타의 fret 간격 비율을 모사하되, 시뮬레이션 상의 물리적 안정성을 고려하여 다음과 같이 설정되었다.

$$S = [0.65, 0.7, 0.8, 0.9, 0.95, 1.1, 1.2, 1.2, 1.3, 1.2] \quad (9)$$

이러한 접근 방식은 고정된 절대 좌표를 사용하는 대신, 양 끝단의 앵커 위치와 비율 벡터 S 만을 조절함으로써 다양한 크기와 형태의 악기 모델에 유연하게 대응할 수 있는 장점을 가진다.

260105유리

위에 작성해준 내용을 수정해봤는데, 이 의도가 맞는지 확인 바랍니다. 위치는 reward function 앞으로 옮겨야 할 것 같네요.

Let $\mathbf{p}_{start}, \mathbf{p}_{end} \in \mathbb{R}^3$ denote the start and end points of a guitar string. The string is conceptually divided into multiple fret regions. To discretize each fret region into $N = 10$ non-uniform intervals, we define the displacement vector along the string as

$$\mathbf{d} = \mathbf{p}_{end} - \mathbf{p}_{start}$$

The intervals within each fret region are assigned non-uniform scaling factors

$$\alpha = [0.065, 0.07, 0.08, 0.09, 0.095, 0.11, 0.12, 0.12, 0.13, 0.12],$$

which approximate the relative spacing between frets on a guitar, while also taking into account the physical stability required for simulation. This approach has the advantage of flexibility: rather than using fixed absolute coordinates, the system can accommodate instruments of different sizes and shapes by adjusting only the end-points and the scaling factors. The corresponding interval vectors are

$$\Delta \mathbf{p}_i = \alpha_i \mathbf{d}, \quad i = 1, \dots, 10.$$

The points along a fret region are then obtained cumulatively as

$$\mathbf{p}_0 = \mathbf{p}_{start}, \quad \mathbf{p}_i = \mathbf{p}_{start} + \sum_{k=1}^i \Delta \mathbf{p}_k, \quad i = 1, \dots, 10,$$

ensuring that the discretized points span exactly from $\mathbf{p}_{start}$ to $\mathbf{p}_{end}$. Finally, the target midpoint M_j for the j -th fret, which represents the center coordinate that the agent should strike, is computed as the cumulative starting position plus half of the corresponding interval vector:

$$M_j = \mathbf{p}_{j-1} + \frac{1}{2} \Delta \mathbf{p}_j, \quad j = 1, \dots, 10.$$

251229 기훈

We use Isaac Gym[] and PyTorch[] for physics simulation and deep learning, respectively.

Table N shows all the parameters we use in our system. For the controller, value function, and the discriminator network, fully-connected feed-forward deep networks of which hidden units are $[n, m]$ are used, respectively, where ReLU activation function is used.

Phycis Sim.	Simulation rate (Hz)	60
PPO	—	—

Table 1. Performance comparison

251226 기훈

(ViSA: Physics-based Virtual Stunt Actors for Ballistic Stunts[Kim et al. 2025] 참고했습니다. 표현을 바꿔야 할 것 같습니다.)

캐릭터의 기본 상태는 Figure 6과 같다. 하체는 앉은 상태로 고정시켜 불필요한 움직임을 최소화하고, 상체는 캐릭터가 기타를 받히고 왼손은 기타의 Neck 위에, 오른손은 기타의 Body위에 자연스럽게 놓아 기타를 받히는 자세이다.

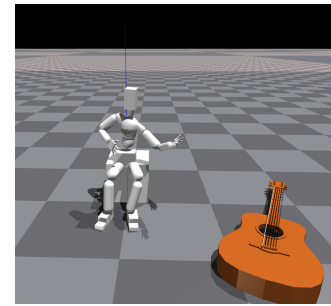


Fig. 6. 251226 기훈 캐릭터의 기본 pose는 다음과 같다.

25/12/31 수빈 Isaac Gym 환경에서 사람과 기타, 그리고 의자의 기본 위치는 Figure 7과 같다. 실제 기타 연주자의 모습을 반영하여, 사람의 허벅지 위에 기타가 안정적으로 위치하도록 했다.

260113 수빈 시계열 음악 데이터(MIDI)를 Isaac Gym 공간 내의 3차원 목표 좌표로 변화하기 위해 다음과 같은 방식을 사용했다.

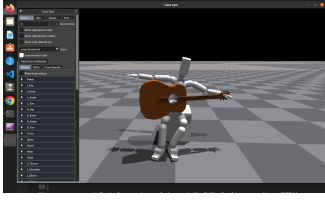


Fig. 7. 25/12/31 수빈 Isaac Gym 환경에서 기타와 사람의 기본 위치

7.2 Exponential Fret Modeling

현악기의 Fretboard 상에서 Pitch와 물리적 거리 사이의 관계를 정의하기 위하여, 등주율에 기반한 지수 모델을 적용한다. 기타의 Nut로부터 n 번째 fret까지의 상대적 거리 비율 R_n 은 다음과 같은 수식으로 표현된다.

$$R_n = 1 - \left(\frac{1}{2}\right)^{\frac{n}{12}} \quad (10)$$

여기서 12는 옥타브 당 반음의 개수를 의미하고, 해당 공식을 통해 이산적인 음악 신호를 지판 위의 정밀한 좌표로 치환 가능하다. 이때, n 은 Nut로부터 fret 번호를 의미한다.

8 Result and Analysis

9 Conclusion

References

- Simon Alexanderson, Rajmund Nagy, Jonas Beskow, and Gustav Eje Henter. 2023. Listen, Denoise, Action! Audio-Driven Motion Synthesis with Diffusion Models. *ACM Transactions on Graphics* 42, 4 (July 2023), 1–20. doi:10.1145/3592458
- Jinseok Bae, Jungdam Won, Donggeun Lim, Cheol-Hui Min, and Young Min Kim. 2023. Pmp: Learning to physically interact with environments using part-wise motion priors. In *ACM SIGGRAPH 2023 Conference Proceedings*. 1–10.
- Valentin Bazarevsky, Ivan Grishchenko, Yurii Kartynnik, George Gunayer, Artem Vakunov, Andrei Tkachenka, Sururoop Mishra, Rohan Khanna, Vladimir Iglovikov, Ruslan Rahaman, Tyler Zhu, Volodymyr Kropytianskyi, and Matthias Grundmann. 2020. BlazePose: On-device Real-time Body Pose tracking. *arXiv preprint arXiv:2006.10204* (2020). <https://arxiv.org/abs/2006.10204>
- Ryan Canales and Sophie Jörg. 2025. Real-time Hand Motion Synthesis for Playing a Virtual Guitar. In *Proceedings of the 2025 18th ACM SIGGRAPH Conference on Motion, Interaction, and Games*. 1–11.
- Jiali Chen, Changjie Fan, Zhimeng Zhang, Gongzheng Li, Zeng Zhao, Zhigang Deng, and Yu Ding. 2021. A music-driven deep generative adversarial model for guzheng playing animation. *IEEE Transactions on Visualization and Computer Graphics* 29, 2 (2021), 1400–1414.
- George ElKoura and Karan Singh. 2003. Handrix: animating the human hand. In *Proceedings of the 2003 ACM SIGGRAPH/Eurographics symposium on Computer animation*. 110–119.
- Katerina Fragkiadaki, Sergey Levine, Panna Felsen, and Jitendra Malik. 2015. Recurrent Network Models for Human Dynamics. *arXiv:1508.00271 [cs.CV]* <https://arxiv.org/abs/1508.00271>
- Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, and Ruoming Pang. 2020. Conformer: Convolution-augmented Transformer for Speech Recognition. *arXiv:2005.08100 [eess.AS]* <https://arxiv.org/abs/2005.08100>
- Mohamed Hassan, Yunrong Guo, Tingwu Wang, Michael Black, Sanja Fidler, and Xue Bin Peng. 2023. Synthesizing Physical Character-Scene Interactions. *arXiv:2302.00883 [cs.GR]* <https://arxiv.org/abs/2302.00883>
- Hsuan-Kai Kao and Li Su. 2020. Temporally guided music-to-body-movement generation. In *Proceedings of the 28th ACM International Conference on Multimedia*. 147–155.
- Christian Kehling, Andreas Männchen, and Arndt Eppler. 2023. *IDMT-SMT-Guitar Dataset*. doi:10.5281/zenodo.7544110
- Minseok Kim, Wonjeong Seo, Sung-Hee Lee, and Jungdam Won. 2025. ViSA: Physics-based Virtual Stunt Actors for Ballistic Stunts. *ACM Trans. Graph.* 44, 4, Article 131 (2025). doi:10.1145/3731424
- Sunmin Lee, Sebastian Starke, Yuting Ye, Jungdam Won, and Alexander Winkler. 2023. Questensim: Environment-aware simulated motion tracking from sparse sensors. In *ACM SIGGRAPH 2023 Conference Proceedings*. 1–9.

- Bochen Li, Akira Maezawa, and Zhiyao Duan. 2018. Skeleton Plays Piano: Online Generation of Pianist Body Movements from MIDI Performance. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*. 218–224.
- Changyang Li and Lap-Fai Yu. 2023. Generating activity snippets by learning human-scene interactions. *ACM Transactions on Graphics (TOG)* 42, 4 (2023), 1–15.
- Ruilong Li, Shan Yang, David A. Ross, and Angjoo Kanazawa. 2021. AI Choreographer: Music Conditioned 3D Dance Generation with AIST++. *arXiv:2101.08779 [cs.CV]* <https://arxiv.org/abs/2101.08779>
- Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. *arXiv preprint arXiv:1906.08172* (2019). <https://arxiv.org/abs/1906.08172>
- Julietta Martinez, Michael J Black, and Javier Romero. 2017. On human motion prediction using recurrent neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2891–2900.
- Josh Merel, Yuval Tassa, Dhruva TB, Sriram Srinivasan, Jay Lemmon, Ziyu Wang, Greg Wayne, and Nicolas Heess. 2017. Learning human behaviors from motion capture by adversarial imitation. *arXiv:1707.02201 [cs.RO]* <https://arxiv.org/abs/1707.02201>
- Hegel Emmanuel Pedroza Villalobos, Termeh Taheri, Wallace Abreu, Ryan Corey, and Iran R. Roman. 2025. *Guitar-TECHS: An Electric Guitar Dataset Covering Techniques, Musical Excerpts, Chords and Scales Using a Diverse Array of Hardware*. doi:10.1109/ICASSP49660.2025.10887996
- Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. 2018. DeepMimic: example-guided deep reinforcement learning of physics-based character skills. *ACM Transactions on Graphics* 37, 4 (July 2018), 1–14. doi:10.1145/3197517.3201311
- Xue Bin Peng, Yunrong Guo, Lina Halper, Sergey Levine, and Sanja Fidler. 2022. ASE: large-scale reusable adversarial skill embeddings for physically simulated characters. *ACM Transactions on Graphics* 41, 4 (July 2022), 1–17. doi:10.1145/3528223.3530110
- Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. 2021. Amp: Adversarial motion priors for stylized physics-based character control. *ACM Transactions on Graphics (TOG)* 40, 4 (2021), 1–20.
- Zhiping Qiu, Yitong Jin, Yuan Wang, Yi Shi, Chao Tan, Chongwu Wang, Xiaobing Li, Feng Yu, Tao Yu, and Qionghai Dai. 2025. ELGAR: Expressive Cello Performance Motion Generation for Audio Rendition. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*. 1–9.
- Javier Romero, Dimitrios Tzionas, and Michael J. Black. 2017. Embodied Hands: Modeling and Capturing Hands and Bodies Together. *ACM Transactions on Graphics, (Proc. SIGGRAPH Asia)* 36, 6 (Nov. 2017).
- Eli Shlizerman, Lucio Dery, Hayden Schoen, and Ira Kemelmacher-Shlizerman. 2018. Audio to body dynamics. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 7574–7583.
- Chen Tessler, Yoni Kasten, Yunrong Guo, Shie Mannor, Gal Chechik, and Xue Bin Peng. 2023. CALM: Conditional Adversarial Latent Models for Directable Virtual Characters. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Proceedings (SIGGRAPH '23)*. ACM, 1–9. doi:10.1145/3588432.3591541
- Guy Tevet, Sigal Raab, Brian Gordon, Yonatan Shafir, Daniel Cohen-Or, and Amit H. Bermano. 2022. Human Motion Diffusion Model. *arXiv:2209.14916 [cs.CV]* <https://arxiv.org/abs/2209.14916>
- Jonathan Tseng, Rodrigo Castellon, and C. Karen Liu. 2022. EDGE: Editable Dance Generation From Music. *arXiv:2211.10658 [cs.SD]* <https://arxiv.org/abs/2211.10658>
- Qingyang Xi, Rachel M. Bittner, Johan Pauwels, Xuzhou Ye, and Juan P. Bello. 2019. *GuitarSet*. doi:10.5281/zenodo.3371780
- Pei Xu and Ruocheng Wang. 2024a. Synchronize Dual Hands for Physics-Based Dexterous Guitar Playing. In *SIGGRAPH Asia 2024 Conference Papers (SA '24)*. ACM, 1–11. doi:10.1145/3680528.3687692
- Pei Xu and Ruocheng Wang. 2024b. Synchronize dual hands for physics-based dexterous guitar playing. In *SIGGRAPH Asia 2024 Conference Papers*. 1–11.
- Yinzen Xu, Weikang Wan, Jialiang Zhang, Haoran Liu, Zikang Shan, Hao Shen, Ruicheng Wang, Haoran Geng, Yijia Weng, Jiayi Chen, Tengyu Liu, Li Yi, and He Wang. 2023. UniDexGrasp: Universal Robotic Dexterous Grasping via Learning Diverse Proposal Generation and Goal-Conditioned Policy. *arXiv:2303.00938 [cs.RO]* <https://arxiv.org/abs/2303.00938>
- Fan Zhang, Valentin Bazarevsky, Artem Vakunov, Andrei Konotopskaia, Mihail Hashavsky, Ivan Grishchenko, Vijit Chu, Yurii Kartynnik, Shagun Enakshi, Artem Ablavatski, and Matthias Grundmann. 2020. MediaPipe Hands: On-device Real-time Hand Tracking. *arXiv preprint arXiv:2006.10270* (2020). <https://arxiv.org/abs/2006.10270>
- Yuanfeng Zhu, Ajay Sundar Ramakrishnan, Bernd Hamann, and Michael Neff. 2013. A system for automatic animation of piano performances. *Computer Animation and Virtual Worlds* 24, 3-4 (2013), 445–457.

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